

Case Study 2

Regression Testing and Change Validation for Previously Certified iGaming Builds

Portfolio case study for regulated iGaming QA, certified build comparison and compliance focused regression testing

Case Study Summary

This case study demonstrates how I validate a new iGaming build against a previously certified version to confirm that approved functionality, game rules, payable values, artwork, financial logic and compliance related content remain unchanged unless the change is authorised.

In regulated iGaming, regression testing is not only about confirming that existing features still work. It is also about confirming that a product still aligns with what was previously approved for certification. A small change in a help screen, payable value, symbol rule, payout description or game behaviour can create certification risk.

Business Problem

When a game has already been certified, any new build must be carefully checked before it can be accepted for further review or release. Even if the change request looks small, the new build may introduce unexpected changes that affect gameplay, rules, payouts, artwork, financial behaviour or player facing information.

A build can fail certification if the tester identifies differences that were not declared, approved or documented.

Business Risk

Risk Area	Possible Impact
Unauthorised changes	New build may differ from the certified version
Incorrect payable values	Player facing payout information may be wrong
Rule disclosure issues	Help screen may not fully explain game behaviour
Financial logic risk	Wagers, wins or payouts may calculate incorrectly
Certification delay	Product may need to be returned for correction
Market entry delay	Operator cannot launch until issues are resolved
Compliance exposure	Regulator may reject the product or request further evidence

Test Objective

The objective of this case study is to verify that a new iGaming build remains aligned with the previously certified version, except for approved and documented changes.

Validation Area	Objective
Build version	Confirm that the correct new build is under test
Certified baseline	Confirm that the previous certified version is used for comparison
Change scope	Confirm that declared changes are understood
Game rules	Confirm that rule descriptions remain accurate
Paytable	Confirm that symbol values and payout values match expected values
Artwork	Confirm that symbols, labels and feature text remain correct
Gameplay	Confirm that base game and feature behaviour still work correctly
Financial logic	Confirm that wagers, wins and balances remain accurate

Validation Area	Objective
Game history	Confirm that round records are still captured correctly
Help screen	Confirm that player facing information is complete and accurate
Source or file changes	Confirm that changes are authorised and expected

Test Scope

In Scope	Description
Build comparison	Compare the new build against the previously certified version
Game launch	Confirm that the game launches successfully
UI regression	Confirm that screens, buttons and labels display correctly
Paytable validation	Compare payout values against the approved version
Help screen validation	Confirm that rules and feature descriptions are complete
Symbol validation	Confirm that symbols match the approved artwork and descriptions
Base game behaviour	Confirm that normal game rounds behave as expected
Feature behaviour	Confirm that bonus or free spin features behave correctly
Transaction validation	Confirm that wagers, wins and balances remain accurate
Game history validation	Confirm that records are created correctly after each round
Defect retesting	Confirm that previously fixed issues remain resolved
Evidence capture	Capture screenshots, comparison tables and test results

Out of Scope

Out of Scope Area	Reason
Full mathematical certification	The case study focuses on regression and change validation
RNG certification	Randomness certification is not part of this sample
Security penetration testing	Security testing is outside the scope
Production data	All data is fictional and recreated for portfolio use
Real client builds	No confidential employer, client or product information is used

Test Environment

Item	Example Value
Application type	Demo iGaming slot game
Certified version	Version 1.0
New build version	Version 1.1
Test environment	QA environment
Browser	Chrome, Edge or Firefox
Test account	Funded test player
Test management	JIRA style defect tracking
Evidence	Screenshots, comparison table, test cases and defect report
Database	MySQL or PostgreSQL

Change Request Summary

For this portfolio example, the fictional change request is:

Change Item	Description
Build update	Game updated from version 1.0 to version 1.1
Approved change	Minor artwork update on loading screen
Approved change	Typo correction on help screen
Not expected	Paytable values should not change
Not expected	Game rules should not change
Not expected	Wager limits should not change

Change Item	Description
Not expected	Payout behaviour should not change

Certified Build Comparison

Test Area	Certified Version 1.0	New Build 1.1	Expected Result
Game title	Demo Treasure Slot	Demo Treasure Slot	Must match
Minimum bet	R1.00	R1.00	Must match
Maximum bet	R100.00	R100.00	Must match
Wild symbol rule	Wild substitutes for all symbols except scatter	Wild rule must remain complete	Must match
Scatter trigger	3 or more scatters trigger free spins	3 or more scatters trigger free spins	Must match
5 high symbols payout	100x	100x	Must match
Loading screen artwork	Old artwork	Updated artwork	Approved change
Help screen typo	Typo present	Typo corrected	Approved change
Game history	Wager, win, balance and round ID displayed	Same details displayed	Must match

Detailed Test Case 1

Test Case ID

TC REG 001

Test Case Title

Verify that payable values remain aligned with the certified version

Test Objective

To confirm that the new game build displays the same approved payable values as the previously certified version.

Preconditions

Precondition	Description
Certified version is available	Tester has access to approved baseline evidence
New build is deployed	Version 1.1 is available in QA
Change request is available	Tester understands approved changes
Test account is funded	Player can launch and interact with the game
Evidence capture is available	Screenshots can be taken for comparison

Test Steps

Step	Action	Expected Result
1	Open the certified version evidence pack	Approved payable values are visible
2	Launch the new game build	Game opens successfully
3	Open the payable screen	Paytable loads correctly
4	Compare symbol names	Symbol names match the certified version
5	Compare symbol images	Symbol images match the certified version unless approved
6	Compare payout values	All payout values match the certified version
7	Compare feature payout descriptions	Feature descriptions match the approved rules
8	Capture screenshot evidence	Evidence is saved for review
9	Record test result	Test is marked as pass if all values match

Expected Result

All payable values in the new build should match the previously certified version. Only approved changes listed in the change request may differ.

Paytable Regression Comparison

Symbol	Certified Version 1.0	New Build 1.1	Result
Crown 5 symbols	100x	100x	Pass
Crown 4 symbols	50x	50x	Pass
Crown 3 symbols	20x	20x	Pass
Gem 5 symbols	80x	80x	Pass
Gem 4 symbols	40x	40x	Pass
Gem 3 symbols	15x	15x	Pass
Wild rule	Substitutes except scatter	Substitutes except scatter	Pass
Scatter trigger	3 plus triggers free spins	3 plus triggers free spins	Pass

Detailed Test Case 2

Test Case ID

TC REG 002

Test Case Title

Verify that help screen rules remain complete after build update

Test Objective

To confirm that all player facing game rules remain complete, clear and consistent with the certified version.

Test Steps

Step	Action	Expected Result
1	Open the certified help screen evidence	Approved rules are visible
2	Launch the new game build	Game opens successfully
3	Open the help screen	Help screen loads successfully
4	Check base game rules	Rules match certified version
5	Check wild symbol rules	Wild behaviour is fully explained
6	Check scatter symbol rules	Trigger condition is clearly explained
7	Check free spin rules	Free spin award and behaviour are explained
8	Check wager information	Minimum and maximum wager information is correct
9	Capture evidence	Screenshots are saved
10	Record result	Test is passed only if all required rules are present

Sample Defect Report

Defect ID DEF REG 001
Defect Title Wild symbol substitution rule missing from help screen in new build
Severity Medium
Priority High
Environment QA Build 1.1
Module Help Screen
Related Test Case TC REG 002

Description

The new build does not display the full wild symbol substitution rule on the help screen. The certified version explains that the wild symbol substitutes for all symbols except scatter. In the new build, the wild symbol is shown visually, but the substitution rule is missing from the text.

Steps to Reproduce

Step	Action
1	Launch the new game build
2	Open the payable screen
3	Locate the wild symbol
4	Open the help screen
5	Review the wild symbol rule text
6	Compare it against the certified version evidence

Expected Result

The help screen should state that the wild symbol substitutes for all symbols except scatter.

Actual Result

The wild symbol appears on the payable, but the help screen does not explain the substitution rule.

Evidence

Evidence	Description
Screenshot 1	Certified version help screen showing wild rule
Screenshot 2	New build help screen missing wild rule
Screenshot 3	Paytable showing wild symbol
Comparison table	Certified version compared with new build
Test case result	TC REG 002 marked as failed

Business Impact

The player may not receive complete information about how the game works. This can create confusion during gameplay and may affect the product's readiness for certification review.

Compliance Risk

Incomplete rule disclosure may be treated as a compliance issue because player facing game information must accurately explain how game features behave.

Recommendation

The help screen text should be updated to include the full wild symbol substitution rule. Once corrected, the build should be retested against the certified version and the evidence pack should be updated.

Regression Test Scenarios

Test Case ID	Scenario	Expected Result
TC REG 001	Paytable values match certified version	All values match
TC REG 002	Help screen rules remain complete	All rules are present and accurate
TC REG 003	Loading screen artwork reflects approved update	Only approved artwork change is present
TC REG 004	Minimum and maximum wager values remain unchanged	Wager limits match certified version
TC REG 005	Base game round completes successfully	Round completes and records correctly
TC REG 006	Free spins trigger condition remains unchanged	Feature triggers as approved
TC REG 007	Game history records wager, win and balance	History values are correct
TC REG 008	Previously fixed defect remains resolved	No regression occurs
TC REG 009	Game interruption recovery remains correct	Round resumes or resolves as expected
TC REG 010	Build version displays correctly	Correct version is shown

Regression Execution Summary

Metric	Count
Total regression test cases	10
Passed	8
Failed	1
Blocked	0
Not run	1
Defects logged	1
High priority defects	1
Critical defects	0

Sample Regression Result

Area	Result	Comment
Build launch	Pass	Game launches successfully
Paytable comparison	Pass	All values match certified version
Help screen validation	Fail	Wild rule missing
Artwork change	Pass	Approved loading screen update only
Wager limits	Pass	Limits unchanged
Base game behavior	Pass	Rounds complete correctly
Free spins behavior	Pass	Trigger condition unchanged
Game history	Pass	Transaction records display correctly

Area	Result	Comment
Defect retest	Pass	Previous issue remains fixed
Final recommendation	Conditional fail	Correction required before approval

Evidence Pack for Portfolio

For the actual portfolio, create a folder called Case Study 2 Certified Build Regression Testing. Inside it, include the following files:

File Name	Purpose
Regression Test Plan.pdf	Explains the test approach and scope
Certified Build Comparison.xlsx	Compares certified version and new build
Regression Test Cases.xlsx	Contains detailed regression test cases
Paytable Comparison.xlsx	Compares payout values
Help Screen Comparison.pdf	Shows recreated help screen evidence
Defect Report.pdf	Shows sample defect documentation
Regression Summary Report.pdf	Shows execution summary and recommendation
README.md	Explains the case study on GitHub

Final QA Recommendation

Based on the executed regression scope, the new build should not proceed to certification approval until the missing wild symbol rule has been corrected and retested.

Although the approved artwork and typo changes were successfully validated, the missing help screen rule introduces a player information and compliance risk. The corrected build should be revalidated against the certified baseline, and the updated evidence should confirm that the help screen, payable and game rules are fully aligned.

Skills Demonstrated

Skill	How It Was Demonstrated
Regression testing	Compared new build behaviour against certified version
Compliance testing	Checked player facing rules and certification related content
Change validation	Confirmed that only approved changes were introduced
Paytable validation	Compared symbol and payout values
UI validation	Checked artwork, labels and help screen content
Defect reporting	Logged a clear defect with business and compliance impact
Risk analysis	Explained why incomplete rules affect certification readiness
Evidence preparation	Captured screenshots, comparison tables and test results
Test documentation	Produced test cases and regression summary
Attention to detail	Identified a small text issue with regulatory impact

Portfolio Disclaimer

All examples in this case study are recreated for portfolio purposes. No confidential employer, client, product, jurisdiction, system or regulatory information is included. The case study is based on real testing principles used in regulated iGaming environments, but all data, screenshots, identifiers and records are fictional.